



Recognition memory decisions made with short- and long-term retrieval

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Abstract

In the present research, we produce a coherent account of the storage and retrieval processes in short- and long-term event memory, and long-term knowledge, that produce response accuracy and response time in a wide variety of conditions in our studies of recognition memory. Two to nine pictures are studied sequentially followed by a target or foil test picture in four conditions used in Nosofsky et al. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 47, 316–342, (2021) and in our new paradigm: VM: target and foil responses to a given stimulus change from trial to trial; CM: the responses do not change from trial to trial; AN: every trial uses new stimuli; MIXED: combinations of VM, CN, and AN occur on each trial. In the new paradigm a given picture is equally often tested as old or new, but only in CM is the response key the same and learnable. Our model has components that have appeared in a variety of prior accounts, including learning and familiarity, but are given support by our demonstration that accuracy and response time data from a large variety of conditions can be predicted by these processes acting together, with parameter values that largely are unchanged. A longer version of this article, containing information not found here due to space, is available online <https://doi.org/10.31234/osf.io/h8msp>. The availability of the data (supplement materials), info and link is attached at the end section (<https://psyarxiv.com/h8msp>).

Keywords Short term and long term memory · Recognition · Learning · Memory models

This research aims to show how the varied processes of storage in and retrieval from both short- and long-term memory work together to allow memory to operate. The goal is not to test individual hypotheses but rather to produce a comprehensive account, to demonstrate that the many processes of memory working together can explain the primary accuracy and response time results from short-term recognition studies. The participant studies a short sequentially presented list of pictures and decides whether a test picture had been on that list ('target'), or had not ('foil'). The lists varied in length from 2 to 9 in the three studies analyzed in this report. There were four primary conditions: 'Varied Mapping' (VM): target and foil responses to a given stimulus change from trial to trial; 'Consistent Mapping' (CM): the responses to a given stimulus do not change from

trial to trial; 'All new' (AN): every trial uses new stimuli; 'MIXED': combinations of VM, CN, and AN occur on each trial. Mixed conditions increase the likelihood that the same memory processes and strategies are used for each type of picture. We also use two forms of CM trials. Each item in the traditional CM paradigm is consistently mapped to both the categories of target or foil and the two response keys. In the new CM paradigm, a given item is equally often tested as a target or foil but is always mapped to the same response key. In total this report analyzes and models accuracy and correct response time data from 376 conditions in 13 groups of participants that were separately tested in VM, AN, two forms of CM, and several conditions mixing VM, AN, and CM.

We account for the data with processes that have separately, but not cohesively, received support in many prior publications. Most prior research was aimed to assess particular hypotheses with data from limited conditions, to support models with limited number of components, or to predict accuracy but not also response time. The present modeling integrates what we believe to be the most important memory processes into a single theory. The success of the model in accounting for accuracy and response time

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results from a wide variety of conditions, with parameter values that mostly are unchanged across all the conditions and groups, gives confidence that the model captures the major processes of memory and the way they work together.

The goal of the present research does not lend itself to the more common approach in which one or two hypotheses are specified and then tested with results that support those hypotheses and reject certain others. Our research should not be judged successful because one can point to individual findings supporting or rejecting one or two hypotheses (although we can do this in one or two instances), but rather because the model accounts well for the entire pattern of results across all conditions and groups. It is important to note that doing so is far from easy: Just one part of the present data set was modeled by Nosofsky et al. (2021). Those authors found it challenging to find a model that could predict their findings with a set of common parameter values: As will be detailed later, their model was fit separately to different sets of their conditions, and despite this, the values of some of the parameters required to produce good predictions did not make sense conceptually. As we will show in this report the present model accounts well for all of the data in that prior publication, as well as the data from new experiments reported here, using a model with parameter values that are largely unchanged across the studies, groups, and conditions—values that we argue make sense conceptually.

The experiments we present are complex with many conditions. Accounting for the results requires a complex model, making it easy to get lost in the details and lose track of the big picture. Thus, we begin with a conceptual overview of the main features of the model.

Overview of the model

Memory traces are stored in long-term memory and short-term memory. These are stored when items are studied and when items are tested. Short-term traces are event traces (also known as episodic traces), containing not just the nominal item studied but also the local and personal context identifying the environmental and personal context in which the event occurs. Long-term traces are of two kinds: Event traces and knowledge traces. Long-term event traces are stored as imperfect copies of short-term event traces. Knowledge traces result from the accumulation of many similar events, thus losing identification with any single context but containing useful information such as the appropriate response to be made to a given stimulus (the meaning of a word seen; where to run to catch a baseball); in our studies, these form in CM conditions.

Retrieval operates by querying memory with a probe containing both content and context information. For short-term recognition testing (as in the present article) the testing probe consists of the current context plus a studied item from

the recent list (OLD, or a ‘target’) or current context plus an item not from the current list (NEW, or a ‘foil’). Everything we have learned and know is stored in knowledge. Thus when a test item is presented, it is encoded (perceived) by retrieving features stored in knowledge, with low-level physical features and later with higher-level, more abstract semantic features. At each moment the current features are used to probe recent event traces to carry out the task of recognition. The most readily available event traces are in short-term memory and those will be activated and retrieved first; activation and retrieval of long-term event traces (e.g., from recent prior lists), will take place at a slower rate. In addition, in CM conditions of a recognition paradigm, the appropriate response for the test item has been learned and stored in a knowledge trace that also contains the item and the task context. The encoded response will be retrieved in parallel with what is being retrieved from event traces and will join the event trace information in helping to produce an appropriate response.

In VM conditions the responses are not consistent from trial to trial: targets on some trials are foils on other trials, and these occur equally often. Thus a knowledge trace formed for a VM item will not have useful information about the appropriate response, and the only useful information for the recognition decision will be that retrieved from event traces. Furthermore, retrieval from trials prior to the current trial generally provides inconsistent information and thus adds noise, harming performance. Because prior trial activation harms performance, participants will try to weight the current context cue heavily, thus focusing retrieval as much as possible on the memory traces of the current list. These traces are assumed to be in short-term memory, but with strength strongest for the most recently studied items. Recent list focusing is imperfect, so activation of prior trial traces, particularly recent prior trials, will occur and harm performance.

Our recognition model assumes that decisions are based on the gradual activation of memory traces by the probe in a process of evidence accumulation (a random walk) until enough evidence is accumulated to reach a decision of target or foil. Trace activation is based on the similarity of the momentary set of probe cues to each trace. Initially, probe cues consist of low-level physical features (e.g., lines, edges, colors) that are assumed to activate only short-term event traces. Subsequently, activations of different types of traces can occur depending on the training condition for the test stimulus: An AN test stimulus activates mostly short-term event traces. A VM test stimulus activates short-term traces and mostly long-term event traces that match the current test item. A CM test stimulus activates short-term traces, recent long-term event traces, and the responses in knowledge traces. For CM test stimuli, our model is unique in assuming parallel activation of all three kinds of traces, all

three combining to produce the activation that governs each step of evidence accumulation leading to a decision.

Activation of short-term and long-term *event* traces is summed to produce a value of ‘familiarity’. This value does not depend on the response that might be encoded in the event traces. The value of familiarity determines a tendency for the random walk to take a step toward a target or foil decision, but that step is determined probabilistically, so event trace activation can result in a momentary tendency for the random walk to move toward the ‘wrong’ decision boundary. In contrast, activation of a knowledge trace (for trials with a CM test stimulus) automatically activates its encoded response, producing a tendency for the random walk to take a step toward that response, but not the alternative response (see Schneider & Shiffrin, 1977; Shiffrin & Schneider, 1977). Thus the step direction produced by knowledge trace activation is not probabilistic—such activation causes the step direction produced by such activation to be toward the correct response. However, step direction results from a combination of activation of knowledge traces and event traces, so overall step direction is probabilistic. Depending on the strength of knowledge trace activation relative to event trace activation, either source can be the more effective in determining the momentary step direction.

Our studies have relatively rapid presentation times and short delays until test. The model (for this type of study) assumes that the familiarity value of a target increases with its recency of presentation in the study list. The implicit assumption is that each successive item joining short-term memory interferes with the previously stored short-term traces, reducing their quality of storage; activation is a product of quality and similarity so activation decreases as short-term traces are older (have larger ‘lags’). This assumption produces a strong recency of target performance.

Finally, the model allows for occasional human error: It is assumed that a small percentage of trials are ‘glitches’ producing random responses at random times.

Background and rationale

With one exception, the model components described above have appeared in many prior publications; the way that learning and familiarity contribute in parallel to evidence accumulation is unique to both the model of Nosofsky et al. (2021), and the present model.

It has long been recognized that most memory tasks require retrieval from both short- and long-term memory (e.g., Atkinson & Shiffrin, 1968, for recall; Mandler et al., 1969, for recall and recognition; Atkinson & Juola, 1974, for recognition). For recognition testing these reports also give evidence for the use of two processes, recall (defined as the use of the contents of memory traces) and

familiarity (defined as summed activation of traces). In this tradition, the present modeling makes use of similar assumptions.

Much of the modeling of memory since the 1950s uses accuracy of retrieval as a primary measure. Particularly for modeling recognition, researchers have produced models that predict both accuracy and response time (e.g., Atkinson & Juola, 1974; Ratcliff, 1978). Response times are often used as a measure for recognition paradigms because response times for binary responses are relatively easy to interpret. Especially in recent years models predicting both accuracy and response time have been based on ‘evidence accumulation’: Probabilistic evidence is accumulated in steps until sufficient evidence favors one response over the other.

It is rare to see research and modeling of short-term recognition based on accuracy, response time, and different combinations of VM, AN, and CM, the exceptions being a good deal of prior research of this sort by Nosofsky and colleagues (Cao et al., 2017, 2018; Donkin & Nosofsky, 2012a, b; Nosofsky, 2011, 2016; Nosofsky, Cao, et al., 2014a; Nosofsky, Cox, et al., 2014b; Nosofsky & Donkin, 2016; Nosofsky et al., 2011). These studies collected, analyzed, and modeled both accuracy and response time. A recent study by Nosofsky et al. (2021) contrasted CM, VM, and AN and added mixed conditions that were not used in the earlier reports.

The new studies presented in this report used VM, AN, and mixed conditions but with the new form of CM described above and detailed shortly. The new studies were designed to clarify the cause of the superiority of CM to VM and AN: In all three conditions, targets have more familiarity than foils because they have traces on the current list that are activated. The CM advantage is presumably due to retrieval from prior trial event traces or knowledge traces. It is natural to assume that response learning occurs in CM and is stored in event traces, knowledge traces, or both. However, in the standard CM paradigm targets are encountered more often than foils as trials continue, because targets are studied and tested but foils are only tested. Thus, CM targets should have higher familiarity than foils due to activation of study traces from the current trial (which is true for VM and AN also) but they should have additional familiarity due to activation of study traces from prior trials. AN has no traces from prior trials for targets or foils and VM has equal numbers of traces from prior trials for both targets and foils. Thus the CM performance superiority could be due to higher familiarity from study traces on prior trials. The new CM conditions equate familiarity from prior trials for targets and foils, so the CM advantage must be due to response learning. We also hoped that the new conditions would help us design an improved model because the modeling in the 2021 paper had certain problems, as described later. Indeed we report an improved

model, one fit to both the new results and those from the 2021 study.

The various conditions in the 2021 study and the present studies highlight the way that both familiarity and recall might be used together to carry out recognition. A good deal of research has shown that it is sufficient to model recognition accuracy with a ‘familiarity’ decision model (e.g., Dunn, 2008). Decisions in such a model are based on the activation of short-term traces and long-term event traces (from prior trials), but not based on responses in those traces. Yet sufficiency does not imply an absence of recall, where recall involves access to the contents of a particular trace including the response to be given. A number of lines of research, such as those using the ‘remember-know’ approach (e.g., Gardiner & Java, 1990), have suggested the use of recall. The senior author is one of many who have suggested that the sufficiency of familiarity is due to redundancy: In many common recognition paradigms the use of recall would usually lead to the same response as the use of familiarity (e.g., Cox & Shiffrin, *in press*). The present model continues to adopt this approach to the use of recall and familiarity: The role of recall is only represented by response learning due to activation of, and presumably recall from knowledge traces. Recall of event traces might well be occurring on some trials, but we do need to add such processes to the model to predict the results.

Method

Overview

The present studies (and the 2021 study) use a probe recognition paradigm with picture stimuli: A short list of pictures is presented one per second for study. The lists vary in length from 2 to 8 (3–9 in Nosofsky et al., 2021). The test picture is one-half the time a target (‘OLD’) and one-half the time a foil (‘NEW’). The participant gives a binary response whose accuracy and response time are measured. AN refers to tests of pictures that are new on every trial of the experiment. VM refers to tests of pictures that are targets on some trials and foils on others. There were two different versions of CM, one used in the standard paradigm and one used in the present studies, as described in detail shortly. Both the 2021 publication and the present one had ‘pure’ and ‘mixed’ conditions for different groups of participants. In pure conditions, all trials were either AN, VM, or CM. In the 2021 study, each trial of the mixed conditions had equal numbers of study pictures of all three types; in the present studies, each trial of the mixed conditions had equal numbers of two of these three types.

The most obvious feature of the new paradigm is the spatial position of the monitor on which the pictures are

presented. In the standard paradigm (used in Nosofsky et al., 2021) the list of pictures to be studied and the test picture both occur in the screen center. In the new paradigm, the study and test pictures occur either on the right side or left side of the screen center. By itself, this should be an unimportant change, and the results for AN and VM should remain largely identical to those in the standard paradigm.

CM conditions overview

There are two variants of CM, one used in the standard paradigm and one used in the present studies. The term CM will be used to describe both because the context will make it clear which is being discussed. In the standard paradigm, CM targets remain targets on every trial and CM foils remain foils on every trial. Targets are both studied and tested but foils are only tested, so targets appear more often than foils. In the present studies, one set of CM pictures is always presented for study on a given side of the screen center, and when tested as a target (i.e., after study on that side) requires a key press on that side. However, that same picture occurs as a foil for trials on the other side of the screen center, and a foil requires an opposite side key press. Thus a CM picture always requires the same key press and that key press response can be learned. The other set of CM pictures is the same, except presented for study on the other side of the screen center, and hence requiring a key press on that side either when tested as a target (on that side) or as a foil (on the other side). Thus, all CM pictures have a consistent key press that can be learned. VM pictures in the new paradigm occur randomly for study and test on both sides, thus requiring inconsistent key presses, but occurring exactly as often as each CM picture. This design equates familiarity for CM and VM, so that the cause of an observed CM superiority must be due to learning, not extra familiarity.

Method details for the present studies

The new design presented the pictures for study on a given trial on either the left side of fixation (left columns in Fig. 1) or the right side of fixation (right columns in Fig. 1). Examples of four CM trials for a participant in this new design are given in the four panels of Fig. 1. A test picture follows, and when the trial was one on the left side a key press on the left side of the keyboard is used to indicate that a target has been tested and a key press on the right is used to indicate that a foil has been tested. On a right-side trial, a key press on the right is used to indicate that a target has been tested and a key press on the left is used to indicate that a foil has been tested. In the CM conditions, there are two sets of *N* pictures, one always used for study lists on the left and the other always used for study lists on the right. In both cases, the foil is from the set on the other side. With this design, a

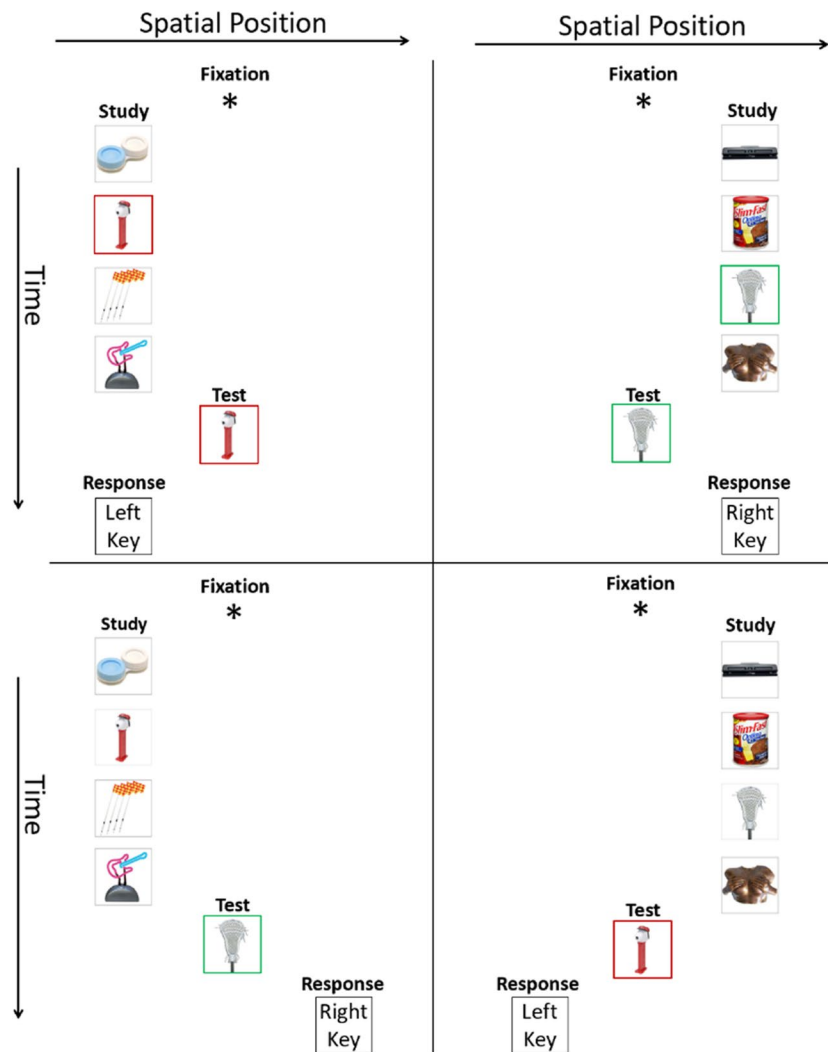


Fig. 1 The top panels show left and right-side target test trials for Set Size 4 that could have occurred in CM, VM, AN, or MIXED conditions. In AN conditions, no picture repeats on any trial. In VM conditions, a given picture is studied randomly on a given side as trials continue. In CM conditions, a picture is always studied on the same side. Thus, the bottom two panels show foil tests for CM that could have occurred on other trials: Because the response key for a target is

on one side of the keyboard, and for a foil is on the opposite side, the comparison of the top and bottom panels show that for CM the same response key is used for the same picture, whether a target or foil. The bottom two panels also show foil tests that could have occurred on some of the VM foil trials, but on other trials the same picture could have been a foil on the other side, so on VM trials the response keys are inconsistently mapped to the pictures. (Color figure online)

given CM picture, whether a target or foil, always requires a response key on the same side, allowing the key press response to be learned. The key press for a given CM stimulus can also be learned in the standard CM procedure, but the participant can also learn a mapping of a given stimulus to the category ‘target’ or the category ‘foil’. In the present design, this category learning is not possible because a given CM stimulus is tested equally often as a target and a foil. A given CM picture will be studied only on a given side, but that fact won’t determine whether that picture when tested is a target or foil. The VM conditions have $2N$ pictures; on each trial, a random subset is selected for study and assigned randomly to a side; in this case, the key press cannot be

learned. AN has new pictures every trial, regardless of side, and of course, learning is not possible.

In the present design (and also the standard paradigm), *short-term trace activation* is larger for targets than foils, in each of CM, VM, and AN, because a tested target but not a foil strongly matches one trace due to its presentation on the most recent list. In the present paradigm, long-term trace activation is the same for targets and foils, not only for VM but also for CM. Thus it is only learning (the key to press) that can produce CM superiority to VM: There is no extra long-term familiarity for CM targets. The two present studies were very similar (as described below) and carried out with about a one-year delay between.

Probe recognition

On each trial, the participant sees a series of pictures to remember, presented one at a time, each in the same location on the computer monitor. Following this study list a single test picture is presented; with probability 0.5, the test picture will be a randomly chosen picture from the study list (called a ‘target’, or OLD); with probability 0.5, the test picture was not on the study list (called a ‘foil’, or NEW). The study lists have 2, 4, or 8 pictures, the number chosen randomly on each trial.

With probability 0.5, the study list and test picture are presented on the left side of the screen center, and otherwise on the right side of the screen center. There are two keyboard letters used to respond: ‘F’ (the key on the left) and ‘J’ (the key on the right). The participant is instructed to respond to a target with the key on the side of the presentation, and to a foil with the other key. Participants are instructed to put index fingers on the two response keys at all times and to respond as fast and as accurately as possible. They are given practice for a brief period at the start of a session. They are not informed of the differences across conditions in stimuli or response mappings (i.e., CM, VM, AN), and are not told the length of the coming list prior to each trial.

Stimuli and apparatus

The stimuli are chosen from 2,400 daily life pictures from the website of Talia Konkle (Brady et al., 2008). Each image subtended a visual angle of approximately 7 degrees and is displayed on the designated side of the screen approximately 20 degrees from the screen center and roughly halfway from bottom to top. The experiment is conducted with MATLAB Psychophysics Toolbox (Brainard, 1997) on PCs. All participants are tested in private, sound-attenuated booths.

Trial timing

On each trial, a fixation point (*) appears on the center of the screen for 0.5 seconds. Each picture in the study list is displayed on the designated side for that trial for 1 s, with an interstimulus interval between pictures of 0.1 second. Following the study list, for Experiment 1, a test picture appears at the screen center; for Experiment 2 a test picture appears at a position halfway from the study pictures to the screen center (and hence still on the side of the presentation). In both experiments, the change of position signals to the participant that a test has occurred, and a response is needed. The test probe stays on the screen until the key response is registered. A blank screen then lasts for 0.5

seconds followed by feedback (*correct* or *incorrect*) for 1 second. Then the next trial starts.

Groups and conditions

Across both experiments, there are nine groups with different participants. Each group has 9 blocks of 25 trials in experiment 1 and 24 trials in experiment 2. The first block is training and not analyzed. The assignment of participants to conditions within each group is counterbalanced. For each block, half the trials use the left side, and half the right side, randomly permuted. The study lists contain either 2, 4, or 8 pictures, these set sizes are randomly mixed in each block.

Pure VM

Used in both Experiments 1 and 2. Each participant has 16 pictures randomly chosen from the 2,400. On each trial, the study list pictures are randomly selected from the 16. A foil is randomly selected from those pictures not on the study list.

Pure CM

Used in both Experiments 1 and 2. Each participant has 16 pictures randomly chosen from the set of 2,400 pictures, 8 assigned to be targets on the left-side trials and foils on the right-side trials, 8 assigned to be targets on the right-side trials and foils on the left-side trials.

Pure AN

Used only in Experiment 2. Each participant has new stimuli randomly chosen from the 2,400 for every trial.

MIX4 (VM, CM)

Used only in Experiment 1. Each trial’s study list has half CM and half VM stimuli. Both VM and CM have eight pictures randomly selected for each participant from the 2,400. Four CM pictures are assigned to be targets on the left and foils on the right, and the other 4 CM pictures are assigned to be targets on the right and foils on the left. The eight VM pictures are randomly assigned to each trial to side and target/foil.

MIX8 (AN, CM)

Used only in Experiment 2. Each trial’s study list has half CM and half AN pictures, the CM pictures consisting of 16 stimuli randomly selected for each participant from the 2,400. Eight CM pictures are assigned to be targets on the left and foils on the right; the other 8 CM pictures are

assigned to be targets on the right and foils on the left. The AN stimuli are new every trial, randomly selected each trial for each participant from the 2,400.

MIX4 (AN, CM)

Used only in Experiment 2. Each trial's study list has half CM and half AN pictures, the CM pictures consisting of 8 stimuli randomly selected for each participant from the 2,400. Four CM pictures are assigned to be targets on the left and foils on the right; the other 4 CM pictures are assigned to be targets on the right and foils on the left. The AN stimuli are new every trial, randomly selected each trial for each participant from the 2,400.

AT

Used only in Experiment 2. Pure VM and Pure CM conditions alternate every trial, each with a different set of 16 stimuli selected randomly for each participant from the 2,400. The results did not match the corresponding results from the other conditions and groups. We judged that the AT participants used idiosyncratic mixtures of strategies. Because we are presenting a coherent account of a large set of results with a model assuming a common strategy and choice of criteria for all groups, conditions, and participants, we do not analyze and model the AT results.

Conditions used in Nosofsky et al. (2021)

This study used the standard design with all study pictures and test pictures presented at fixation: The five conditions were: Pure AN; Pure VM; Pure CM; and MIXED (VM, CM, and AN with equal numbers in each study list). However, the MIXED condition was run twice at different times—we use MIX1 to refer to Cohort 1 and MIX2 to refer to Cohort 2. Thus, there were six groups in all. Eighteen pictures were used for the pure VM conditions. The pure CM conditions used 9 targets and 9 foils. In the mixed conditions, the VM pictures were chosen from a set of 6 pictures, and the CM targets and foils were chosen from sets of three each. Details may be found in Nosofsky et al. (2021).

Participants

Participants were undergraduate students at Indiana University who received course credits as requirements in an introductory psychology course. They gave consent and were allowed to leave at any time without penalty. The study and participation were approved by the Indiana University IRB.

There were 89 participants in total in Experiment 1. Two were deleted from the analysis for obvious failures to obey instructions. The numbers in each condition were

thus 28 (CM), 28 (VM), 31(MIX4 (VM, CM)). There were 204 participants in total in Experiment 2. Two participants were deleted from the analysis for obvious failures to obey instructions. The numbers in each condition were thus 34 (CM), 35 (VM), 33 (AN), 32 (MIX8 (AN, CM)), 34 (MIX4 (AN, CM)), and 34 (AT). The experiment took about 45 minutes for each participant to complete. There were 194 participants in Nosofsky et al., (2021).

Results

Because we will use the same model to predict the results from our new studies and those from the corresponding conditions of Nosofsky et al. (2021), we show in all figures results (and predictions) for the conditions of both. Trials to be excluded were decided qualitatively, based on observations of the distributions of correct response times, in advance of other analyses, and in advance of modeling of the data. In Nosofsky et al. (2021), data from trials where responses occurred before 150 ms or after 4,000 ms were excluded (~1% of the data). In the current more complicated study, response times were more irregular, and more data were excluded: Data from trials where responses occurred before 180 ms or after 1,300 ms were excluded (~18% of the data in Experiment 1 and ~15% of data in Experiment 2). Given these decisions were made in advance of analysis and modeling, they were not biased.¹

It is a daunting task to find a good way to display the accuracy and correct response time results from a very large number of conditions.² The 8 groups (and different conditions) of the present studies and the 5 groups (and different conditions) from Nosofsky et al. (2021) comprise 376 distinguishable conditions/data points for accuracy and another 376 for correct response time medians. Figure 3 presents all the results, with target lag on the horizontal axis, different panels for different conditions, and different list lengths in each panel. This method of graphing makes it hard to compare results from condition to condition. Figure 2 aggregates over target lag, and places list length (set size) on the

¹ In response to requests from a reviewer and editor, the present data were analyzed using the 2021 criteria, but median RTs became very irregular and noisy. The present data were then analyzed with criteria of 180 and 2,000; this dropped the excluded data from about 16% to 6%; this shift changed accuracy imperceptibly and caused median RT to become slightly more noisy but without changing any of the trends. The original chosen criteria are used for the present data because they were chosen in advance and hence are unbiased.

² An Appendix in the archived report gives the values and number of observations for every condition, and for correct response times of the average of the medians, and their standard deviation, across the participants. That appendix also gives the model predictions for each point.

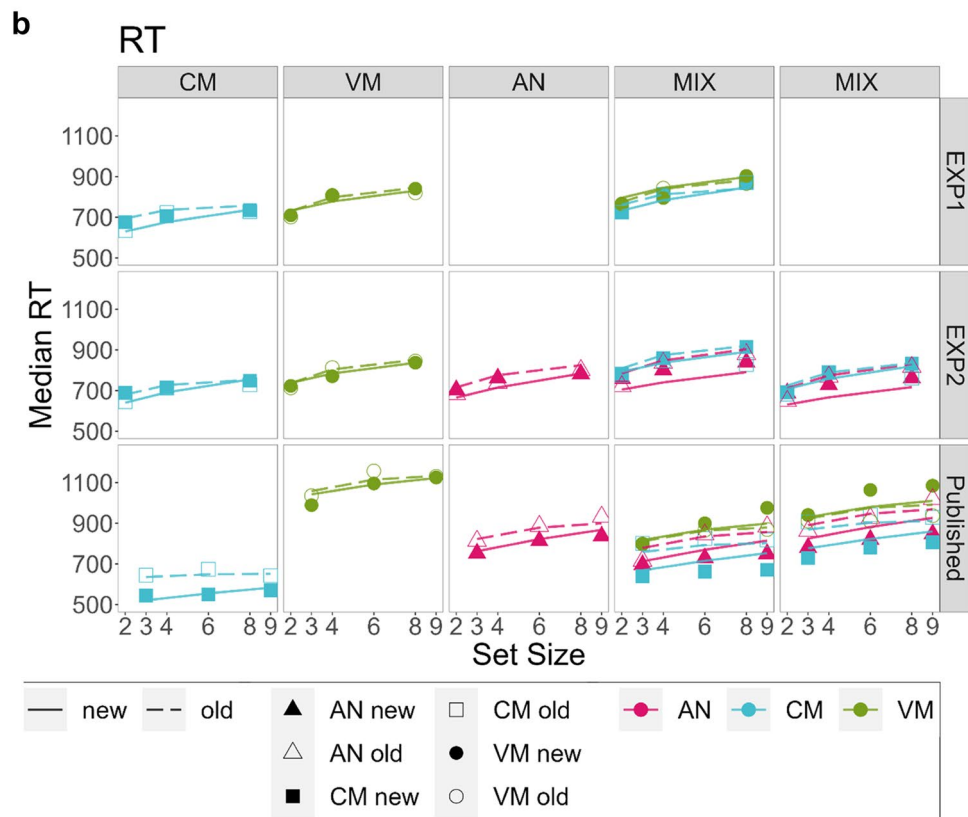
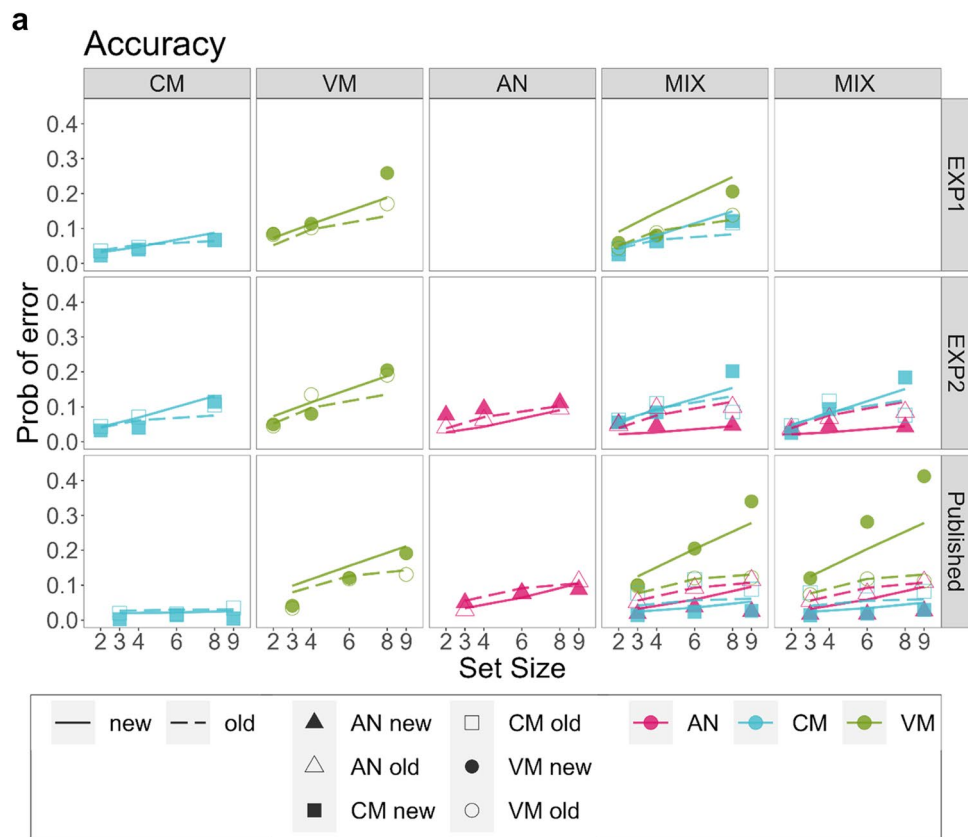


Fig. 2 Observed data (symbols) and predictions (lines) as a function of set size. Open symbols and dashed lines are targets (OLD) and filled symbols and solid lines are foils (NEW). **a** Accuracy. **b** Median correct response times. The mixed conditions in the last two columns are organized as follows: (1) Row 1: Experiment 1, Column 4: MIX4 (VM, CM). (2) Row 2: Experiment 2; Column 4: MIX4 (AN, CM); Column 5: MIX8 (AN, CM). (3) Row 3, Nosofsky et al.; Column 4: MIX9 (AN, CM, VM), Cohort 1. Column 5: MIX9 (AN, CM, VM), Cohort 2. (Color figure online)

horizontal axis. Various combinations of AN, VM, and CM are in each panel. This display method makes it easier to see the main findings. In order not to replicate these figures, the predictions of the model are given on each figure, but those can be ignored for now.

CM versus VM

One aim of the present studies was a demonstration that learning does occur in CM, and that not all the CM performance advantage over VM was due to a larger familiarity advantage for targets over foils. That conclusion was reached by Nosofsky et al. (2021) on the basis of model comparisons, but problems with that modeling (to be discussed later in this report) make it useful to provide a clear empirical demonstration. Figure 2 contains results for CM and VM, for both pure and mixed conditions, and makes it easy to see that the superiority of CM over VM is very large for the data from Nosofsky et al. (2021), and is present but less strong for the present studies. The advantage of CM over VM is most meaningful in the mixed conditions where the same strategy is likely used for storage and retrieval of the CM and VM pictures that are mixed in the same list (although the model uses the same processes for all conditions). The CM advantages in the present study must be due to learning because the paradigm rules out other possibilities. Conservative sign tests show CM superiority in all conditions, particularly including the mixed conditions: Combining hits and correct rejections, the two-sided sign test between VM and CM for combined data of MIX1 (CM, VM) and MIX2 (CM, VM) for the results from Nosofsky et al. (2021) is $p < .0001$ for both correct RT and accuracy. The two-sided sign test between VM and CM for the mixed conditions for the present experiment MIX4 (VM, CM) is $p = .004$ for accuracy and $p = .009$ for correct RT.

The results in Fig. 2 show a much larger superiority for CM over VM for Nosofsky et al. (2021) than for the present study. Some of that superiority could be due to extra familiarity for targets over foils that is only present in the former study. We believe that slower learning is the most important factor accounting for the smaller CM advantage in the present studies. Our modeling (described later) shows that it is sufficient to assume that the difference in the degree of CM superiority is due to slower learning in

the present studies. (Later in this report we have a discussion of the different learning rates in the model).

What is learned in the new paradigm is a mapping of each picture to a given response key. It is possible that such learning occurs directly due to the consistent training of the association. Alternatively, learning could occur in a two-stage process in which the participant learns first that a given stimulus is a target on one side and a foil on the other and then learns a mapping for this picture as a target to a given response key and foil to the other response key, but the opposite mapping for other pictures. The direct model is simpler but it is possible that the slow rate of learning in the new paradigm is due in part to an attempt to learn in two stages.

Set-size effects

Figure 2 makes it easy to see that longer lists have lower performance for accuracy (both hits and correct rejections) and response time for correct responses, for each of AN, CM, and VM. Except for floor effects for the pure CM conditions in Nosofsky et al. (2021), the lower performance for longer lists can be seen as obviously rising error and RT functions in every panel. These results are very strong and reliable, obviating the need for statistics. It is particularly significant that the foil set size effect is so large, reliable, and robust because the large target set size result shown in Fig. 2 is almost entirely due to target recency (termed target lag—see Fig. 3 and the discussion below).

AN versus VM versus CM

For the pure conditions in both studies, performance is best for CM, next best for AN, and worse for VM, as can be seen by comparing panels of Fig. 2 (and Fig. 3). These results are large and reliable, again obviating the need for statistics. For the mixed conditions, the same pattern occurs but the differences are much smaller largely due to extra noise caused by fewer observations per each data point. As discussed when the model is presented, AN is superior to VM because all AN trials have new pictures so an AN test picture has no matching long-term event traces. VM test pictures produce activation of many matching long-term event traces from prior trials that are non-diagnostic with respect to whether the current test picture is a target or foil, so their activation adds variability (often termed noise) and reduces performance. For both VM and AN, there could be some minor activation of non-matching long-term event traces but that effect is very small, and in any event, is roughly the same for tests of AN and VM pictures.

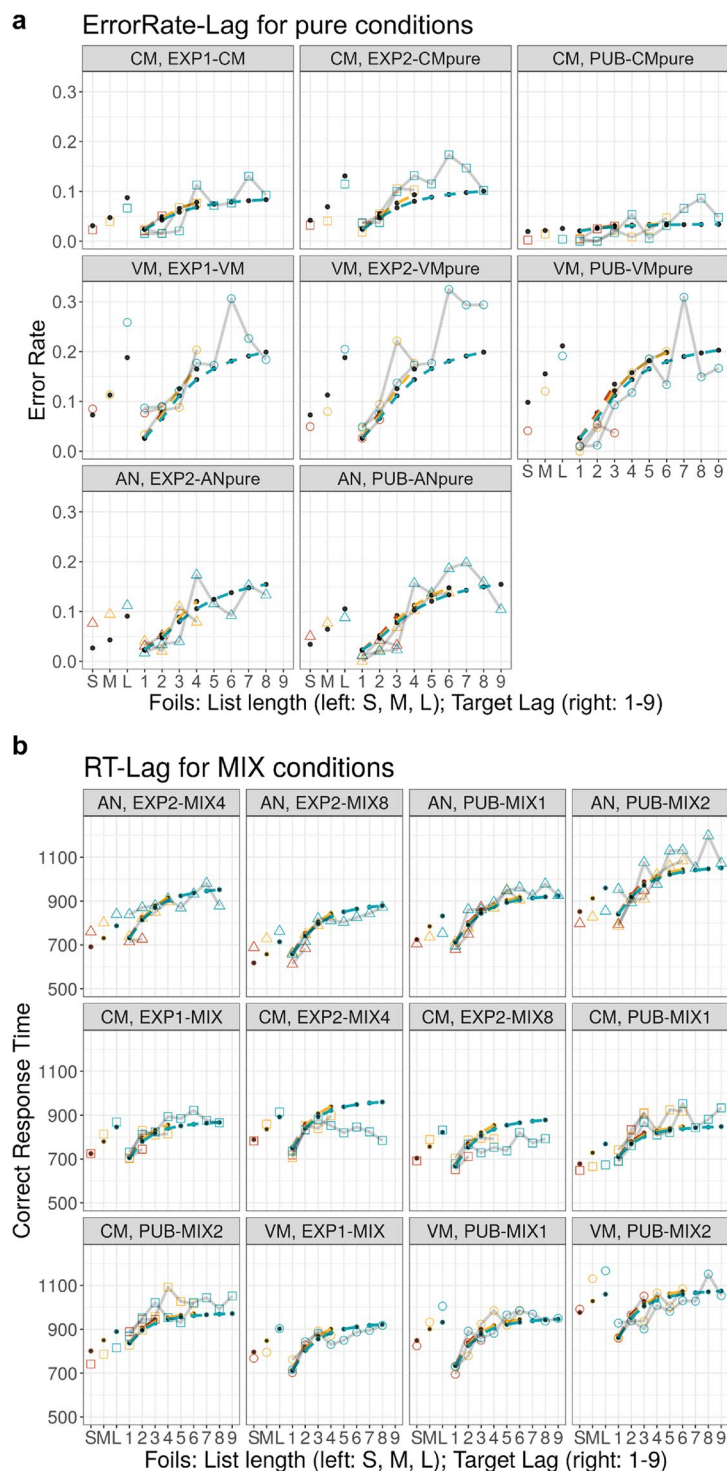


Fig. 3 Observed data and predictions as a function of lag. **a, b** Median correct response times for all of the pure, and mix conditions. **c, d** Accuracy for all of the pure, and mix conditions. PUB refers to Nosofsky et al. (2021); EXP1 and EXP2 refer to present studies. EXP1-MIX refers to MIX4 (CM, VM), EXP2-MIX4 refers to MIX4 (CM, AN), and EXP2-MIX8 refers to MIX8 (CM, AN). Each panel shows results for different list lengths. The lag results for targets for the three list lengths are superimposed on the right portion of each panel, the longest length being on the right end. There is a single foil result for each of the three list lengths; these are on the left of

each panel, with L indicating the longest list length, M the intermediate length, and S the shortest length. Triangles are used for AN, squares for CM, and circles for VM. Observations are the colored and shaped points, connected by the semi-transparent gray lines. Red points/shapes are used for list lengths 2 or 3, yellow points/shapes for lengths 4 or 6, and blue points/shapes for lengths 8 or 9 (the target lags for different list lengths lie more or less atop each other). Predictions are the black dots – for the lags these black dots are connected by colored lines to make the predictions easier to view. (Color figure online)

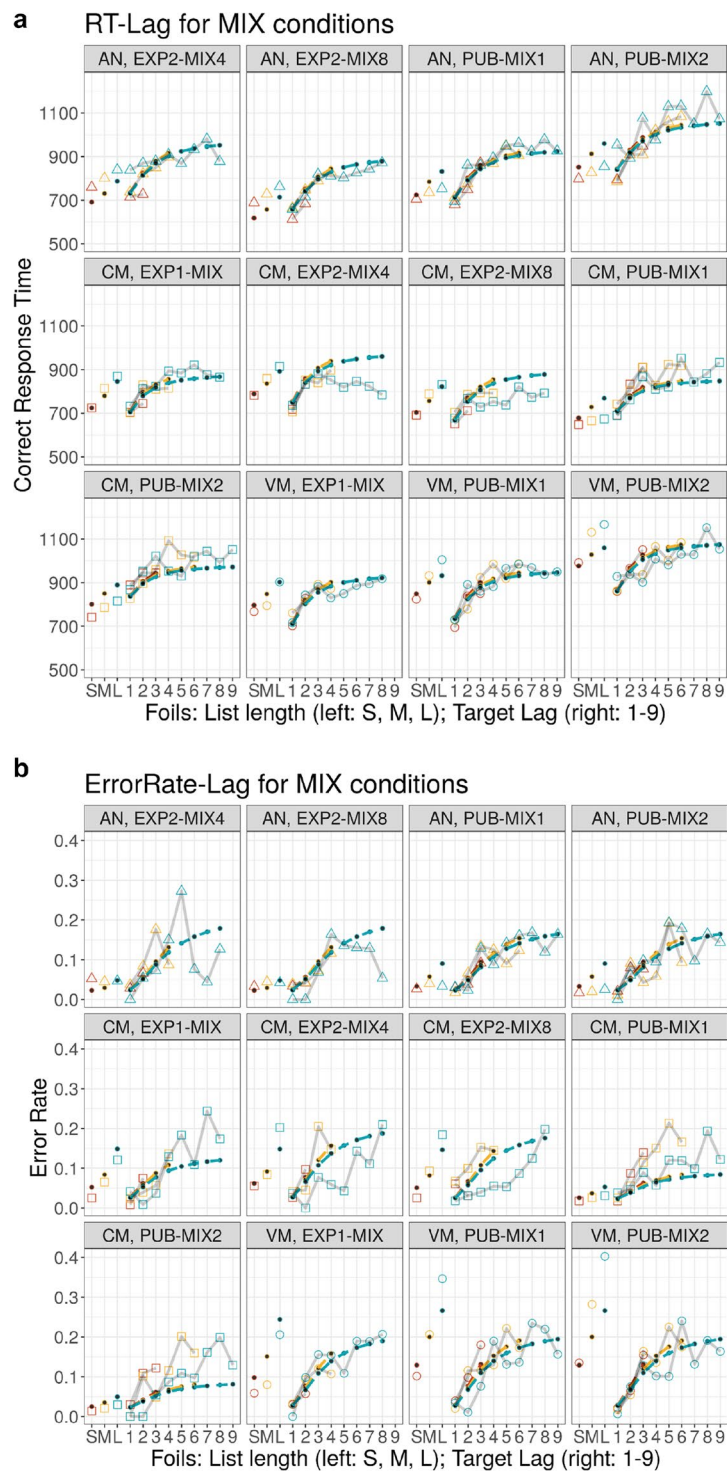


Fig. 3 (continued)

Mixed/Pure

The CM advantage over VM is naturally less for the mixed conditions: Learning is slowed because CM study and test

items occur one-third or one-half as often as in the pure CM conditions. Beyond this result, the mixed results are ‘noisier’ due to fewer observations per data point. We next discuss the effects of target lag, as shown in Fig. 3.

Target lag effects

Figure 3 makes it clear that targets exhibit pronounced recency: The most recently presented items are the most accurate and faster. These results are strong and appear universally, again obviating the need for statistics. Such results are found not just in these reports but in many other studies by Nosofsky and colleagues, and others, leading Donkin and Nosofsky (2012a, b) to propose a general power law model. Recency represents a decrease in the quality of the short-term traces as the age of those traces increases. We do not model explicitly the cause of the lag effects but other lines of research suggest that retroactive interference is the cause, that each item presented decreases the quality of short-term traces stored earlier (e.g., Oberauer et al., 2016).

Target lag effects at different set sizes

It is obvious upon inspection of Fig. 3 that the lag functions tend to lie atop one another for different set sizes, for both accuracy and response time, when aligned to the most recent item. Thus most of the target set size effects shown in Fig. 2, while quite real, are due to the target lag effect: The decrease in performance at longer lags means that *average* performance on longer lists will be lower. At a given lag there is too much noise in the data to produce any significant difference in the target lag results as a function of list length (although the model presented later predicts a very small decrease in performance for longer lists at a given lag). This result could mislead one into thinking performance at a given lag is almost independent of list length. However, recognition performance is determined by target and foil performance together (for example, using signal detection-based measures like d'). There are, in almost all panels of Fig. 3, large and systematic effects of set size for foils (except in a few cases of floor effects): foil accuracy is worse and foil response time is slower for longer lists. Thus the data universally show worse overall recognition performance at each lag for longer lists. That being said, the target-foil ‘asymmetry’ is interesting and potentially significant: A target at a given lag differs little as list length increases, but foils (which are independent of lag) differ a lot. Recognition tasks require a criterion for deciding OLD or NEW, and one could imagine that the criterion is set in accord with the average lag for the current list, but that would produce poorer performance at a given lag for longer lists and would not align with the results. However, if a participant should use the same criterion for all conditions and list lengths, then it might be chosen for the shortest length, or perhaps according to the most recent item at Lag 1. That would produce the lack of a large list-length effect for a target at a given lag, but a large effect list-length effect for foils.

The effects of CM, VM, AN, list length, and target lag on both accuracy and response time are very large and reliable in both the present study and that of Nosofsky et al. (2021). There have been many studies revealing such effects when only a few conditions have been varied, or when only accuracy or response time has been measured. Although the main results shown here are not novel, the present study is unique in its scope (as is the modeling of the results). In addition to the importance of collecting data from a large number of conditions, these findings are important for modeling (i.e., explaining the processes used to carry out short-term recognition). Our model will be laid out shortly, but looking ahead: (1) The modeling captures the main results, results that appear reliably in the many conditions in several studies. This should support the inference that our modeling is capturing reliable trends and important processes of recognition. (2) The fact that the results are consistent with prior research supports the inference that our model captures basic and general processes of memory. (3) Showing consistency of the results from our new paradigm with those from the standard paradigm justifies fitting both with a common model and attempting to do so with parameter values that vary as little as possible across conditions.

The consistency of these major results notwithstanding, their number and relation to each other makes it challenging to produce a coherent model. In fact, the model in Nosofsky et al. (2021) was fit to just a part of the overall data set we present and model in this paper, and although predicting well quantitatively was deficient in several important respects, as will be covered in detail later. Thus the success of the model we present in this paper is to be judged not by the prediction of these major effects (it will do so), but by (a) the ability to predict the relation of the many patterns that are seen; (b) predicting well with a consistent and coherent set of assumptions, each of which have individually received support in prior research; (c) predicting well with parameters that as far as possible have values that do not change across the many conditions and groups; (d) predicting well with parameter values that are sensible and justifiable. No prior research has put together all the many processes of short- and long-term storage and retrieval and shown that they can together explain accuracy and response-time results from such a large array of conditions, let alone satisfying these desirable constraints for assessing the success of the modeling exercise. We turn to the modeling next.

Modeling

The primary goal of the present research was an understanding of the main processes of short- and long-term memory that operate in probe recognition. Studies with some of the present conditions were modeled with good quantitative success by Nosofsky et al. (2021). That model

was based on a number of processes that we believe are important and valid. However, that model had a number of unsatisfactory properties that undermined support for the model's processes. Our new model predicts the data from the earlier study as well as those from the new studies, resolves the ambiguity about the cause of learning, and resolves the problematic issues associated with the 2021 model. This section of the paper will present our model. A later section will explain the problems with the earlier model, highlight the processes that are different and the same, and discuss how the present model resolves the problems of the earlier model.

The new model is termed SLR23 (standing for **STM LTM Retrieval 2023**). The model in the 2021 paper is similarly termed SLR21. SLR23 accounts for the patterns of findings from both Nosofsky et al. (2021) and the present studies, using a common model for both. Due to the large number of groups and conditions, and the large amount of data, the critical information from which we infer the processes at work lies not in statistically significant differences of some data points from others, but rather in the ability of a relatively simple and coherent model to explain the entire set of data patterns for both accuracy and response times. With the partial exception of SLR21, nothing of this scope has been attempted before.

SLR21 used a variant of the EBRW model originally used for categorization experiments by Nosofsky and Palmieri (1997). Because we are fitting both the 2021 data and our new data with a single model, we again use the EBRW model framework. A note on terminology: SLR23 incorporates many of the same processes as SLR21 but differs in ways that make it confusing to use the same notation. Thus SLR23 uses conventional notation, with parameters notated by Greek letters presented in ***bold italic***.

The model we have developed was outlined in conceptual terms at the beginning of this report. This section will describe in detail the way that those conceptual ideas were implemented in the EBRW modeling framework to allow parameters to be estimated and produce a set of predictions for the entire data set. Before turning to the technical details, we make a few remarks concerning the two basic processes used in the present modeling: *learning* and *familiarity*. Learning is a matter of storing and using the response associated with a given item, which can occur in CM because the mapping is consistent across trials, but not in VM because the mapping is completely inconsistent, and not in AN because every trial has new stimuli. One might think learning could be implemented in either of two ways: (1) Responses could be stored in long-term memory in event traces (i.e., episodic memory; Gardiner & Java, 1990; Raaijmakers & Shiffrin, 1980, 1981). (2) Responses could be stored in long-term memory in knowledge traces (i.e., semantic memory; as described in Nelson & Shiffrin, 2013).

Either type of storage, when retrieved, could help govern the next step of evidence accumulation.

SLR23 uses the assumption that learning is due to storage and retrieval only from knowledge traces. The most compelling reason for using this assumption lies in the effects of successive trials. Especially in conditions mixing CM and VM in the same list, if responses were to be stored and retrieved in event traces one would expect that to happen for both the CM and VM items in that list. However, Nosofsky, Cao, et al. (2014a) examined VM and CM cases where the same item occurred as a foil in two successive trials. In VM (but not CM) the foil response on the second trial was much less accurate and much slower than that on the prior trial, or that when a different item occurred on the prior trial. If the response on the prior trial had been retrieved, that should have helped rather than harmed performance on the following trial with the same item and the same response. Thus learning in SLR23 is assumed due to retrieval from a knowledge trace, and the worse performance on the second of two successive VM trials with foils is assumed due to the familiarity process, as described next.

The 'familiarity' process is based on the activation of both short-term event traces (e.g., Gillund & Shiffrin, 1984; Lin & Oberauer, 2022; Malmberg, 2008; Nosofsky, 2016), and the activation of long-term event traces (e.g., Cox & Shiffrin, 2017; Dunn, 2008; Shiffrin & Steyvers, 1997, 1998). Activation of both types of event traces is based on the similarity of the trace to the probe of memory, where similarity is based both on the quality/strength of the trace (e.g., the number of features in the trace) and the degree to which the trace features match and mismatch the probe features. *SLR23 is based on the assumption that the probe does not contain a particular response so that activation is not dependent on the response in an event trace.* Thus, the amount of trace activation is the same whether the event traces encode a target or foil response. The sum of both types of event trace activation helps determine the next step of evidence accumulation, through a sampling process as described soon. To repeat: the familiarity process is based on the activation of event traces by probes containing item and context but not responses, and the amount of activation of each event trace is determined by a match of the item and context in the probe to the limited amount of item and context in the trace. This type of familiarity has been used on many occasions to model long-term recognition; among other findings, it helps explain list length effects (e.g., Gillund & Shiffrin, 1984; Shiffrin & Steyvers, 1997, 1998) and the harm caused by earlier testing of other items (e.g., Criss & Shiffrin, 2004; Criss et al., 2011).

The production of familiarity in SLR23 but not in SLR 21 occurs in two stages as illustrated in Fig. 4, the idea being that the short-term traces are accessed faster than the long-term traces. The need for evidence accumulation in stages

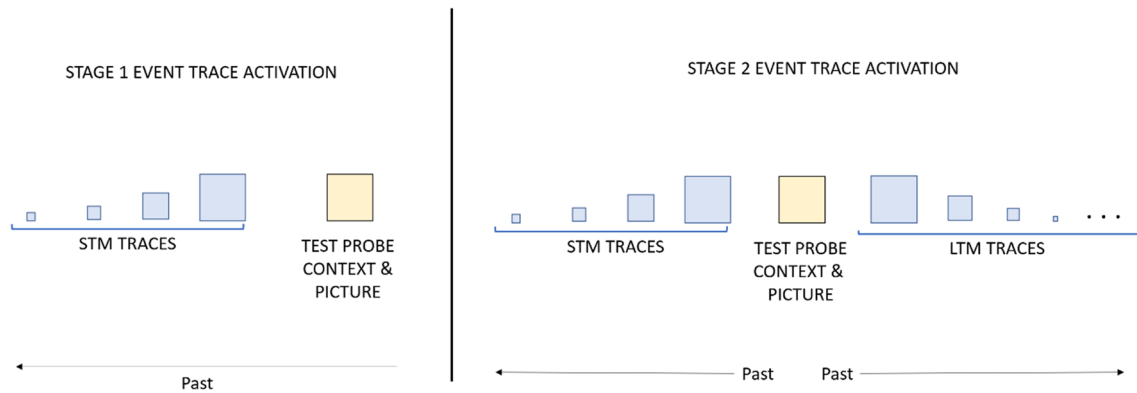


Fig. 4 Familiarity is calculated in two stages, the first summing activation of short-term traces only, and the second summing activations of both short- and long-term traces. The squares descending in size into the past represent the decreasing activation of older traces. (Color figure online)

has received strong support for recognition memory by Cox and Shiffrin (2017) and for perception and decision by Harding et al. (2021).

SLR23 leaves open the question of whether or not responses are stored in event traces, although we believe such storage is likely, based on results from ‘remember-know’ studies. Participants in recognition studies on some trials have the subjective impression of recalling the occurrence of a test item on the most recent list (if and when such recalls occur, they might also include a response). That has led researchers to ask participants to judge whether they ‘know’ or ‘remember’ an item (‘remember’ involves recalling details of the storage event; e.g., Gardiner & Java 1990). Because event trace recall is likely to occur on trials with high test item familiarity it is hard to know if judgments of ‘remember’ are based on recall or high familiarity. For this reason, we make no inference concerning the degree to which event trace recall occurs during recognition testing. Even if responses are sometimes stored in event traces, such responses will not affect the calculation of familiarity because the probe of memory does not include either response. Thus we argue that familiarity is sufficient to explain recognition data. This argument is supported by state-trace analysis (Dunn, 2008). For this reason, we make no inference concerning the degree to which event trace recall occurs during recognition testing.

Implementation of SLR23 in EBRW:

$$P_{TS}(T) = \frac{S_{kT} + L_T + \Lambda_j}{S_{kT} + 2L_T + \Lambda_j + \Phi} \quad (1a)$$

$$P_{TS}(F) = \frac{S_{kF} + L_F + \Lambda_j}{S_{kF} + 2L_F + \Lambda_j + \Phi} \quad (1b)$$

$P_{TS}(T)$ gives the probability of a step toward the target decision boundary (Θ_T) when a target is tested. These steps take place in a process of evidence accumulation characterized as a random walk. Similarly, $p_{TS}(F)$ gives the probability of a step toward the foil decision boundary (Θ_F) when a foil is tested.

S_{kT} gives the summed amount of activation of short-term traces when a target is tested (k is the list-length); S_{kF} gives the summed amount of activation of short-term traces when a foil is tested (k is again the list-length). S_{kT} tends to be higher than S_{kF} because a target tends to match well the trace formed from its presentation on the list whereas a foil does not tend to match well any list trace. Both S_{kT} and S_{kF} rise with the list length k .

L_T gives the summed activation of long-term event traces (from prior trials) when a target is tested and L_F gives the summed activation of long-term event traces (from prior trials) when a foil is tested. $L_T = L_F$ in all conditions except CM in the standard paradigm. Their values vary with condition as detailed later. They do not vary with the current trial list length. They are based on long-term event traces that match the current trial test picture. Long-term traces that mismatch the current trial test picture are activated very little because they mismatch both the content and context in the probe. It is therefore a reasonable approximation to assume they contribute zero activation.

S_{kT} plus L_T , and S_{kF} plus L_F , give the respective target and foil activation that determine the familiarity component of the model.

Λ_j is the amount of activation of the test item’s knowledge trace (that encodes the appropriate response) for the j -th CM condition, and determines the learning component of the model. In Eq. 1a, this produces a tendency to step toward the target boundary and in Eq. 2a produces a tendency to step toward the foil boundary. The degree of learning is allowed to vary across the CM conditions for

what we argue are compelling reasons laid out later when presenting the model's estimated values of Λ_j .

Φ is a criterion determining the direction of the current step in relation to the summed activation values, in the process of evidence accumulation. This criterion was kept constant across all groups and conditions. To see how this operates, ignore the L terms for the moment: Then when Φ equals the value of the numerator, the probability of a step toward the target boundary is 0.5; when larger than the numerator, the step probability is below .5, and when smaller than the numerator, the step probability is above .5. Thus, Φ can be viewed as a criterion.

The use of L in the numerator and 2L in the denominator is used to represent the assumption that there are different roles for short-term event trace activation and long-term event trace activation, due to the fact that long-term traces are from prior trials. Large values of such activation should not increase the tendency to step toward the target boundary but rather should add variability because trace activation from prior trials is assumed to be nondiagnostic. Thus, a higher value of L/2L tends to move the step probability toward 0.5, as it should. Because learning and familiarity are in the same equations, these forces jointly determine the step direction on each trial.

In the rest of this section, traces that were stored due to study or test of the current test picture are termed 'matching' and traces that were stored due to study or test of a picture different from the current test picture are termed 'mismatching'.

Special cases of Eqs. 1a and 1b apply to AN and VM:

For tests of AN targets or foils, there are no long-term matching traces and no learning:

$$P_{TS}(T) = \frac{S_{kT}}{S_{kT} + \Phi} \quad (2a)$$

$$P_{TS}(F) = \frac{S_{kF}}{S_{kF} + \Phi} \quad (2b)$$

As S_{kT} gets larger, and S_{kF} gets smaller, there is a higher probability of getting to a boundary correctly, and doing so faster, so the difference $S_{kT} - S_{kF}$ can be a kind of measure of accuracy. S_{kT} larger than S_{kF} due to the test picture matching a short-term trace, so short-term recognition performance is always above chance.

For tests of VM targets or foils, there is no learning; $L_T = L_F$ because each picture is randomly made a target or foil as trials continue. Thus both are denoted L in Eqs. 3a–b. L/2L moves the step probability toward 0.5, thereby making VM worse than AN (which has no long-term event trace activation).

$$P_{TS}(T) = \frac{S_{kT} + L}{S_{kT} + 2L + \Phi} \quad (3a)$$

$$P_{TS}(F) = \frac{S_{kF} + L}{S_{kF} + 2L + \Phi} \quad (3b)$$

For tests of CM, Eqs. 1a and 1b is used because both familiarity and learning contribute to evidence accumulation. For target tests, Λ_j is in both the numerator and denominator, so increases the step probability toward the correct target boundary. For foil tests, Λ_j is only in the denominator, so increases the step probability toward the correct foil boundary. We next give reasons for the assumptions of the model, and the way they are implemented in Eqs. 1a and 1b.

Evidence accumulation Equations 1a–1b incorporate two kinds of evidence that accumulate, one based on learning that occurs in CM, represented by the Λ term in the equations, and the other based on familiarity, represented by the S and L terms in the equations for short and long-term, that operates in all conditions. Evidence accumulation models have probably been used more than 1,000 times since the 1960s (e.g., Laming, 1968; Link & Heath, 1975; Ratcliff, 1978; Vickers, 1979). Such models are most often used to explain data with both accuracy and response-time measures. These have innumerable forms but have been highly successful, and the present use needs no elaborate justification. In SLR23, evidence starts at zero. A unit step is taken either toward a target boundary or a foil boundary, and this continues until one of these boundaries is reached. There are many problematic issues when one wishes to predict details of the shape of the response time distributions. The present data has too little data to try to predict aspects of the shape, so we predict only medians (medians because means are too sensitive to extreme responses that are outside the model). The use of medians also bypasses most of the problematic issues concerning response time distribution shapes, but this article is not the proper place to go into those issues.

Recency for target tests The quality of traces in short-term memory from the study list decreases with Lag h , where Lag 1 is the most recent item presented (probably due to retroactive interference). This assumption predicts the strong recency effects seen in the target lag functions in many studies using rapid presentations (i.e., Monsell, 1978; Nosofsky et al., 2021). Studies by Sternberg (e.g., Sternberg, 1966) obtained different results, but used slower rates of presentation and test, allowing rehearsal and other strategies to distort the recency effects. Let m_h be the quality/strength of the trace at Lag h :

$$m_h = h^{-\beta} + \alpha. \quad (4)$$

However, the activation of the short-term trace is also dependent on the similarity Ω of the probe to the trace. Conceptually similarity should be determined by the number and value of the content and context features in the probe that match or mismatch corresponding features in the trace. Shiffrin and Steyvers (1997) and Cox and Shiffrin (2017) are two of many who have made such an approach explicit. In the present modeling that approach is implicit because the similarity is an estimated parameter. Thus activation of the trace at Lag h by test item e_i is termed a_{ih} and is a product of its quality m_h and similarity Ω .

$$a_{ih} = m_h \Omega. \quad (5)$$

S_k is then the summed activation of all short-term traces for list length k :

$$S_k = \sum_h a_{ih} \quad (6)$$

Stages of processing The idea that perception operates in a series of stages, with simpler features perceived first, goes back over 100 years. A recent implementation of this idea used for memory retrieval in recognition (and recall) is found in Cox and Shiffrin (2017; see also ; Harding et al., 2021; Holmes et al., 2016). We approximate stages of feature processing with just two stages, the first lasting for just one step of the random walk process. In Stage 1, it is assumed that (1) the test probe uses only low-level physical features, and (2) only the traces in short-term memory are activated. As a result, in Stage 1 AN, VM, and CM items are activated similarly. In Stage 2, high-level features join the test probe and both short- and long-term traces are activated. Stage 2 continues until a response boundary is reached.

Similarity differences governing short-term activation Similarity differences in the activation of long-term event traces are not taken into account because only matching event traces from prior trials are assumed to produce significant activation. We of course want short-term matching traces in both stages of evidence accumulation to have higher similarity than short-term mismatching traces. The similarity of all short-term matching traces in Stage 2 is set to the same value. Doing this is not a meaningful assumption because what matters is relative differences among probe-trace combinations, so setting matching similarity values equal just sets the scale. Thus what does matter is similarity differences in short-term activation when probe and trace mismatch. The implicit assumption is that the high-level features of pictures in different conditions are given different attention and perceived differently, thereby changing the way that probe and trace features match and mismatch. Explicitly this idea is captured by estimating for each condition/combination the degree of similarity Ω , when probe and short-term trace mismatch. The six similarity values that are

Table 1 Probabilities that the test item has a match on the prior trial (SIDE denotes the present experiment and PUB denotes the study of Nosofsky et al., 2021)

	P_T^*		P_F^*	
	$P_{T(E)}^*$	$P_{T(U)}^*$	$P_{F(E)}^*$	$P_{F(U)}^*$
SIDE VM PURE:	1/16	7/24	1/16	7/24
SIDE CM PURE:	1/16	7/24	1/16	7/24
SIDE MIX8-CM/AN CM:	1/32	7/48	1/32	7/48
SIDE MIX4-CM/AN CM:	1/16	7/24	1/16	7/24
SIDE MIX4-CM/VM CM:	1/16	7/24	1/16	7/24
SIDE MIX4-CM/VM VM:	1/16	7/24	1/16	7/24
PUB VM PURE:	1/18	1/3	1/18	1/3
PUB CM PURE:	1/18	2/3	1/18	0
PUB MIX VM:	1/18	1/3	1/18	1/3
PUB MIX CM:	1/18	2/3	1/18	0

needed are listed as part of Table 2. These six values were the same for all conditions, depending only on the picture types (AN, VM, CM) stored during study and used as a test probe. Is it reasonable to allow these similarity values to differ? We return to that question when we discuss the values of the parameters.

Context change and long-term event trace activation Only long-term event traces that match the test item are activated, and the degree of activation is determined by the quality and similarity of the match. One factor contributing to the match is the match of context in the probe to the context stored in the trace. As trials pass, context changes so that activation decreases as the number of trials between trace and test increases. As a simple approximation, we calculate the *probability of a match on the prior trial*, termed P^* , and let total activation be proportional to that probability. These probabilities are not estimates. They vary with group and condition as determined by the design of the studies, with values shown in Table 1. In the column headers, the subscripts denote these conditions: T refers to the test of a target, F to the test of a foil, E to a trace formed due to test, and U to a trace formed by list study.

For a target test, the probability in Table 1 is the sum of the first two columns and denoted P_T^* ; for a foil test the probability is the sum of the last two columns and denoted P_F^* . Activation of long-term event traces are proportional to P^* :

$$L_T = \pi P_T^* \quad (7a)$$

$$L_F = \pi P_F^* \quad (7b)$$

The value π controls the amount of long-term trace activation and therefore represents the weight given to long-term

Table 2 Estimated parameter values

Parameter	Explanation	Specific condition	Estimated value
<i>Parameters that do not change with conditions</i>			
Φ	Criterion		0.54
OLD	The old boundary of the random walk		3.6
NEW	The new boundary of the random walk		-1.5
χ	Response time (ms) per random walk step		91
ξ	Probability of making glitches		0.04
T_ξ	Time in ms. for making glitches		1399
π	The amount of long-term trace activation		0.52
α	Short-term trace quality at Lag 1		0.21
β	Rate of decrease in short-term trace quality		1.70
Ω_2	Short-term similarity between matching test and study pictures in Stage 2		4.1
Ω_{1X}	Short-term similarity between non-matching test and study pictures in Stage 1		1.3
Ω_1	Short-term Similarity between matching test and study pictures in Stage 1		9.4
<i>Parameters that change with conditions</i>			
Ω_{2XX}	Short-term similarity between traces and nonmatching test pictures in Stage 2	VM-VM	0.08
		AN-AN	0.07
		CM-CM	0.13
		AN-VM	0.09
		CM-VM	0.12
		CM-AN	0.03
		MIX-E1	0.33
		CM-E1	0.65
		CM-E2	0.44
		MIX4	0.09
		MIX8	0.03
		MIX1-PUB	0.14
		MIX2-PUB	0.18
Λ	Long-term learning in CM	CM-PUB	1.23
		MIX-E1	133
		CM-E1	107
		CM-E2	95
		VM-E1	108
		VM-E2	112
		MIX4	154
		MIX8	116
		AN-E2	118
		MIX1-PUB	144
		MIX2-PUB	210
		CM-PUB	94
		AN-PUB	164
τ	Base times	VM-PUB	291

trace activation in comparison to short-term trace activation. The value of π is constant across studies, groups, and conditions.

Response times and response boundaries The boundaries for the random walk are θ_T and θ_F for a target and foil responses respectively. These values were fixed across

all conditions. Equation 8 gives the expressions for the response time for the random walk process. The predicted time includes a ‘base time’, or ‘residual time’ τ that represents time for processes not explicitly modeled such as the motor responses required to press the response key. The base times were allowed to vary for the 13 participant groups.

$$RT_{RW} = (N_s)(\chi) + \tau \quad (8)$$

In Eq. 8, N_s is the number of steps needed to reach the boundary, and χ is the time per step.

The subscript RW indicates this is the predicted response time for the random walk. The mean value of N_s for various conditions was determined by an approximation derived by Busemeyer (1982).

Glitches It is assumed that performance is a mixture of ‘regular’ trials and ‘guessing trials’ termed ‘glitches’ (e.g., Clark & Merfeld, 2021; Ratcliff & Kang, 2021). The probability of a glitch is ξ with response time τ_ξ , with values fixed across groups and conditions. Thus:

$$P(C) = (\xi)(0.5) + (1 - \xi)[p(c)] \quad (9)$$

where $P(C)$ is the total predicted accuracy and $p(c)$ is the accuracy predicted by the random walk. The mean response time is:

$$RT = (\xi)(\tau_\xi) + (1 - \xi)RT_{RW} \quad (10)$$

Fitting the data

Julia language is used for modeling (Bezanson et al., 2017), and nonlinear solver IPOPT with JuMP is used for optimization (Dunning et al., 2017). The values shown in Table 2 are close to the best-fitting values but may not be exact due to the limitations of the optimization program that was used. The values minimized a weighted discrepancy between predictions and observations summed across all distinct data points in both studies (those in Figs. 2a–d). The discrepancy in accuracy was a squared difference in probabilities, and the discrepancy in correct response time was a squared difference in seconds between predicted mean time and observed mean of medians across participants (mean of medians was used to eliminate distortions caused by occasional extremely long response times). However, to produce qualitatively good results the discrepancies in accuracy were weighted two times more than discrepancies in correct response time. The discrepancy for foils was weighted four times more than the discrepancy at a given lag (because there is much more data in most foil data points than in most target lag data points). It is worth emphasizing that the only parameter values allowed to vary across groups and conditions were learning rates in the eight CM conditions and base times in the 13 participant groups. There are many other measures of fit that could have been employed, but our aim was not a model producing a precise quantitative account but a model that captures the main processes at work by predicting the data well qualitatively, by predicting the patterns in the data. Thus the predictions we show might have been improved

(slightly) quantitatively by a different measure of fit, but we did not regard that as a consequential goal.

Predictions

The main goal of this research is to show how the most important processes of short- and long-term memory, ones given support in prior research, work together to predict the entire pattern of outcomes of the present experiments, measured both by accuracy and median correct response times. We show next that SLR23 does a good job of capturing the trends in data from a wide array of conditions, groups, and several studies carried out over several years, and does so with parameter values that are largely unchanged with conditions. Thus SLR23 provides an excellent *qualitative* account of the results from both studies. We were pleasantly surprised that the predictions were also reasonably close *quantitatively*.

This can be seen from the predictions superimposed on the data in Figs. 2 and 3. The data that the model fits are of course a mixture of data produced by different people at different times carrying different short-term memory tasks, producing patterns that surely are not identical, and might well vary with different conditions. There is of course no possibility of producing quantitatively accurate fits of such a mixture with a single model using mostly identical parameter values. The hope is that the participants and tasks share the most important basic memory processes and that as a result, the mixture data measured as average accuracy and median response times will reflect those processes well enough to allow the relatively simple model to produce the observed trends. On this basis we judge the model successful: It does capture the data trends qualitatively, as shown by the predictions in Figs. 2 and 3, although missing quantitatively in places. We next describe how the model accounts for the basic trends in the data, trends discussed in the results section of this report.

Recognition is above chance

A tested target tends to match well its short-term trace formed when it was just studied and tends to match poorly short-term traces of other items; a tested foil tends to match all short-term traces poorly. This idea is implemented by similarity differences. The quality of traces decreases with the age of the trace, but identically for target and foil tests. Activation of a given short-term trace is the product of quality times similarity, and thus tends to be higher for target traces. If the list has n short-term traces, then the total short-term activation for a target has a sum of activations for $n - 1$ traces that mismatch, and one that matches; the total activation for a foil test is a sum of $n - 1$ trace activations, all of which mismatch. The step

probability in Equation 1 is largely determined by short-term activation and hence the probability of a step toward the target boundary tends to be higher for a target than a foil for each of AN, VM, and CM.

List length

The extra short-term activation for a target is determined by one trace out of n in the list, so the relative advantage of target activation is diluted by the $n - 1$ mismatches for both targets and foils as n increases, and performance decreases as n increases. Whether this effect is seen for hits (targets), false alarms (foils), or both is determined by criteria settings: the boundaries and the parameter Φ . These values do not vary with list length (or any conditions) and are chosen to match the findings, that a comparison of a target at a given lag to a foil, as length increases, will have the largest effect on foil false alarms and not much on target hits. Why does the data show this pattern? We suspect it is due to each participant receiving many lists of each length. If a participant finds it convenient to use the same criteria for all lists and all target positions, they might choose the values to give reasonable and roughly equal results for the lags that appear at all lengths, lags one (or two or three). Doing so would produce the observed findings, and hence the criteria values in the model.

Learning

Learning is posited to produce the performance advantage of CM over VM and AN. CM leads to the storage of item-response associations in knowledge. These are retrieved in Stage 2 of retrieval and join the familiarity process in determining the direction of the next step of evidence accumulation. Familiarity produces a probabilistic step that might or might not go in the direction indicated by the total short- and long-term event trace activation. Retrieval from the knowledge trace produces a tendency to step toward the correct boundary, but the degree to which that happens is governed by a parameter Λ that varies according to the CM condition of the test item. This is evident in Eqs. 1a and 1b: 1a applies when a CM target is tested, and Λ is in both the numerator and denominator, producing a tendency to move toward the target boundary, and 1b applies when a CM foil is tested and Λ is only in the denominator, producing a tendency to move toward the foil boundary. How strong is either tendency is determined by the strength of Λ relative to the summed strength of short-term and long-term event trace activation (i.e., to the S and L terms in the equations. We will discuss the different estimated values of Λ in a section shortly.

CM versus VM versus AN

CM is superior to VM and AN due to learning, as discussed in the previous paragraph. AN has no knowledge traces because every trial uses new pictures, and VM does not lead to consistent storage of a given response to a given picture, because the response changes half the time. AN is superior to VM due to the activation of long-term event traces in VM—these do not exist in AN. These long-term event traces in VM add noise to the decision processes because they are ‘nondiagnostic’, having the same average value regardless of whether a target or foil is tested on the current trial. That idea is implemented in Equations 1 by the L in the numerator and $2L$ in the denominator: The higher is L the more the step probability tends to move toward 0.5. The model ignores the activation of non-matching test traces, but such traces mismatch the test probe in both content and context, so such activation should be very small. As an approximation such activation is assumed to be missing in the modeling.

Similarity differences in short-term activation

At the level of low-level features like lines edges, and colors, it is unlikely that the similarity of pictures differs depending on whether the test item and trace are of CM, VM, or AN, so the model assumes Stage 1 of retrieval does not activate short-term traces differentially, except that matching traces are activated more than mismatching traces. However, it is quite possible that higher level coding of stimuli depends on the task demands, and task experience, so the high-level features used for storage and probe are different for CM, VM, and AN stimuli. This idea was an important component of the very successful exemplar model (GCM) that Nosofsky has used on numerous occasions for modeling studies of categorization (e.g., starting with Nosofsky, 1986; see also Nosofsky, 2011). We do not have an explicit account of the way similarity should change depending on the type of stimulus tested and the type of short-term trace, except that Stage 1 and Stage 2 matches should have higher similarity than mismatches. For Stage 2 activation of short-term traces that mismatch, the model allows similarity to vary depending on whether the test probe and trace are VM–VM, CM–CM, or AN–AN, and also in the mixed conditions whether the two are VM–CM, VM–AN, or CM–AN. The similarity values were estimated freely for these cases.

A note on similarity differences in short-term activation: We could not find an acceptable model in which all parameter values were fixed across all AN, VM, CM conditions and studies. It is an open question whether another model of equal or better quality could be found that does not assume short-term mismatching similarity values to differ, or that equates various of the differences that we allowed to differ. The present modeling has the merit that the values of

similarity are applied consistently across all studies and conditions, depending only on whether a given probe and trace are VM, CM, or AN.

Model assessment

As can be seen in Figs. 2 and 3, the model captures all the trends in the data in both accuracy and median correct response time. We could not expect and did not intend a good quantitative fit because the data in this study is from different studies at different times, with different groups of participants and many different conditions, surely causing the data to be the result of a mixture of different processes and with a mixture of parameter values where the processes are the same. That being said, there are relatively few large quantitative mispredictions. Two would be problems for any model: The target lag response time results for Experiment 2 MIX4 do not slow as lag increases, which is an exception to what is seen everywhere else. Also, foil data from the two supposedly identical cohorts in Nosofsky et al. (2021) differed in significant ways. We do not regard it as worthwhile to dwell on a few quantitative mispredictions when the model accounts for the qualitative trends quite well. Qualitative is in the minds of the observers, so different observers might come to different assessments of model ‘goodness’. Our judgment might be biased by the difficulty Nosofsky et al. (2021) found in producing an acceptable model for just part of the data, and the several years we worked to develop the present model.

Fit of the data trends is not the only way to assess the quality of the model. Another is the justification for the assumed processes based on a history of prior research, another is related to parsimony and flexibility, and another is the reasonableness of the parameter values used to produce the predictions. The processes of the model were all introduced and given some support in prior research, as discussed earlier. The fact that the model succeeds as well as it does gives support to the proposition that it captures properly the most important processes that are used in short-term recognition tasks: the processes of storage and retrieval in short-term memory, in long-term event memory, and in long-term knowledge. Finally, we argue that SLR23 has justifiable assumptions and justifiable parameter values. We turn to the issue of parameter variations and values next.

Model flexibility, and parameter variations

A plausible case can be made that each of θ_T , θ_F , Φ , ξ , Ω , π , Λ , and τ should vary with participants, groups, conditions, and list length, and these very likely do vary. Such a model would be far too complex and flexible. We searched for a high-quality model with a small number of plausible variations. The variations we allowed were for the base times τ for

the 13 groups, the learning rates Λ in the 8 CM conditions, and the 6 short-term mismatch similarity parameters, Ω_{2XX} . The remaining 18 parameters had values fixed for all groups and conditions.

For a data set this large with this many groups and conditions, with both measures of accuracy and response times, we regard the complexity of SLR22 as relatively low.

Estimated parameter values

To make the case that the present model provides a reasonable and justifiable account of the core processes at work in these studies, it is important that those parameters whose values are interpretable theoretically have reasonable values, particularly in relation to each other.

Short-term decay

The value of β , the short-term decay rate, was about 1.7 and produced the predicted lag curves that conform closely to those observed. The estimated value of α , the asymptotic value of short-term decay, was about .21. This value is hard to interpret, presumably representing some combination of a flattening of short-term loss at long lags, and a contribution from long-term memory.

Learning in CM conditions

The model represents learning as an increase in the probability of taking a step toward the correct response boundary. This representation was used because it had been used in the EBRW modeling in prior publications by Nosofsky and colleagues using similar designs. (We investigated an alternative in which learning is represented by a probability of recalling a correct response, thereby bypassing the usual familiarity process. This mixture model produced predictions similar to those we have shown in this article.) The estimated learning rates had reasonable values. In Nosofsky et al. (2021), both the assignment of pictures to target or foil and to the response key were consistently mapped. The present study used a more complex design, and only the response key was consistently mapped. Thus, the learning rate should have been slower in the new design and was: 1.23 (CM-PUB) > .65 (CM-E1) and .44 (CM-E2); .14 (MIX1-PUB) and .18 (MIX2-PUB) > .09 (MIX4) and .03 (MIX8). In all studies, learning should have been slowed in the mixed conditions relative to the pure conditions because only one-third or one-half of the pictures on each trial were consistently mapped. The pure conditions in fact had faster learning: 1.23 (CM-PUB) > .14 (MIX1-PUB) and .18 (MIX2-PUB); .65 (CM-E1) and .44 (CM-E2) > .33 (MIX-E1), .09 (MIX4) and .03 (MIX8). We argued earlier that learning might have been faster in MIX4 than in MIX8; the

estimates were in this direction but the effect was small: .09 (MIX4) > .03 (MIX8).

Similarity differences between AN, VM, and CM pictures

The similarity of mismatches between probe and short-term traces was allowed to vary depending on whether the probe and stored event trace were different combinations of AN, VM, and CM. We have made a case that these could and should differ, but we had no strong a priori predictions of the way they should do so, other than the prediction that match similarity should be higher than mismatch, which was the case: Stage 1: 9.4 (similarity matching) > 1.3 (similarity mismatching); Stage 2: 4.1 (similarity matching) > {.08, .07, .13, .09, .12, .03} (similarity mismatching for different pairings of probe and studied items). We could not find a model for which all parameters were the same for all conditions, and these similarity differences were one of the primary consequential factors (putting aside base time differences for different groups) that allowed the model to capture the trends in all the conditions. We include the additional discussion of similarity variations in the following section that compares SLR23 to SLR21.

Glitches

The probability of glitch occurrence should be low and was: (.04). Whether glitches are needed at all is a judgment call. Removing glitches while keeping the rest of the model intact, and refitting parameters, produced a somewhat worse fit with somewhat less reasonable parameter values, and with predicted trends slightly less aligned with the data. Whether these issues were sufficient to drop glitches from the model is unclear. In reality, glitches might be needed more to model extreme aspects of the response time distributions rather than medians. We kept glitches in the model in good part because it seems certain at least some participants had some occasions such as lapses of attention that would cause glitches, and other research has identified their presence (or at least showed model improvement when used).

Base times

In evidence accumulation models generally, base times have been used as an adjustment for processes the theorist does not want to model explicitly (such as the propensity to push the response keys more or less quickly), so we followed standard practice in allowing these to vary. Was it reasonable or unreasonable to do so for the 13 participant groups? Had the base times instead been fixed at a single value, but all other parameter values were left unchanged, the accuracy predictions (which were quite good) would not have changed. In addition, the predicted patterns of response

times would have been qualitatively accurate, save for a shift upward or downward of the predictions in each of the response time panels. Although we had no strong a priori predictions, the base time estimates seem generally reasonable, with the most difficult conditions showing long base times. For example, the most difficult condition, VM in the pure condition of Nosofsky et al. (2021) had by far the largest base time (291 ms). The base times seem to be mostly ordered by task difficulty for the 13 groups.

The remaining parameter values are hard to interpret, mostly because they emerge from the interacting processes in EBRW. It should be noted that the parameter space has a number of local minima with similar goodness of fit, with parameter magnitudes varying, but almost always maintaining the ordinal properties listed above.

Model comparison

There are uncountable numbers of model types and variants that could conceivably be employed in attempts to fit the present data set. We believe that a significantly different model of comparable quality would be very difficult to attain, due to the many groups and conditions, and the complexity of the results (and our difficulty in finding the present model). The present project is unique in its scope and prediction of both accuracy and response times; the only exception was the model published by Nosofsky et al. (2021), which modeled some of the conditions also modeled in this report, but was problematic in several respects, as described next.

The model of Nosofsky et al. (2021)

We have termed the model of Nosofsky et al. (2021) as SLR21. It was used to predict the data from five groups (a subset of the 13 groups modeled in this article) and did so quite well. SLR21 had the merit of providing a good quantitative account of a complex data set, used many of the same processes as SLR23, and led to many of the same conclusions. We were led to seek a better model because the way SLR21 was implemented with parameter choices and differences was unduly flexible, and the conclusions reached concerning short-term recognition processes were undercut by estimated values of key parameters that were not reasonable.

We begin by describing briefly SLR21 and the way it was implemented. SLR21 was based on many of the same critical memory processes as SLR 23, such as learning of picture–response associations for CM, and familiarity-based decisions for recognition. The implementation in EBRW was different in subtle but important ways. The basic equation:

$$p_i = (A_i + LTM_{Old}[i] / [A_i + LTM_{Old}[i]]) + (C + LTM_{New}[i]) \quad (10)$$

In this equation, A_i is the short-term trace activation, and learning and long-term trace activation are both handled by the LTM_{Old} and LTM_{New} terms. The simplicity of this equation belies its use in practice: Different parameter values were used to produce predictions for the pure and mixed conditions, and some of the differences in values were quite large, without justification. To put values on these terms required 16 parameters for the mixed conditions and 19 parameters for the pure conditions. In addition, a special parameter ('boost') was added to the mixed parameter group and a different special parameter ('primacy') was added to the pure parameter group. AN was given four different short-term parameters for the mixed conditions but not the pure conditions. Finally, the LTM_{Old} and LTM_{New} terms were allowed to be different for AN, VM, and CM, in effect allowing each of the panels in Fig. 3 for the five groups in Nosofsky et al. (2021) to be fit somewhat independently. Many observers taking these parameter choices and differences into account would not find the resultant good fit to the data impressive.

Putting aside the flexibility of SLR2, the values of the parameters required to fit the data were not plausible:

- 1) In the mixed condition, there was a three times larger learning rate for NEW items than the combined learning and familiarity strength for OLD items, even though old CM items were seen more often and should have been more familiar and learned better.
- 2) The pure CM condition required a huge learning strength for NEW test items and a very small combined strength for OLD test items—again, the opposite of expectations.
- 3) In the mixed conditions, there were equal contributions of long-term traces in VM and AN (equal for both targets and foils), but AN has no matching long-term traces while VM has many.

All these problems were resolved or at the least improved in SLR23: We have presented a case in the previous section of this article that the SLR23 parameters and their values were reasonable, justifiable, and consistent with prior research. Concerning model flexibility, we note that SLR23 used the same equations with the same values to predict together the five participant groups in the pure and mixed conditions of Nosofsky et al. (2021), and also the nine participant groups in the new paradigm of the present studies.

It is difficult to identify precisely the elements of SLR23 that allowed the problems of SLR21 to be resolved. There are some processes in SLR23 not found in SLR21. One was 'glitches', but that change is likely not very important because glitches have a low probability of occurrence and when we removed glitches from SLR23 the main cost was a lowering of the quantitative fit, but the trends in the data

were still predicted. A more important addition was the stages of processing. When we tried removing the first stage from SLR23, while retaining the other processes in that model, several serious problems arose. Unfortunately, pinpointing what stages actually do to improve modeling is difficult because (1) some other high-quality model might be found that does not require stages, and (2) fitting the model without stages leads to changes in the values of other parameters that compensate for the absence of the first stage—such compensatory changes make it difficult to know exactly the sources of the problems. Both SLR21 and SLR23 have complex parameter spaces, interactions, and trade-offs of parameters and processes. Only a few of the parameters (such as the ones governing the loss of quality of short-term traces as successive pictures are presented) have a clear and direct connection to the form of predictions. Nonetheless SLR23 with its processes and estimated parameter values is a 'proof by example' that it is possible to put together the many processes of short and long-term storage and retrieval that operate in multiple conditions of short-term recognition, and account for a wide variety of conditions that measure accuracy and median response times, and do so with largely unchanging parameter values.

We noted that SLR23 resolved most of the problematic issues of SLR21. The word 'most' is used because SLR23 does allow different values for the learning rates, base times, and short-term mismatch similarity parameters. Learning rates were expected to vary, and did so, as expected. It seems reasonable to let base times vary by group. However, the short-term mismatch similarity assumption deserves some additional discussion. The basic assumption is that experience in the tasks leads participants to encode AN, VM, and CM pictures differently, those differences changing similarity between probes and traces that mismatch. Thus three different short-term mismatch similarity parameter values are used for the pure conditions in all studies, for AN–AN, VM–VM, and CM–CM. These same similarity values apply in the mixed conditions of all studies, but in the mixed conditions, three more short-term mismatch values are needed, for AN–CM, AN–VM, and VM–CM. (Long-term mismatching traces are assumed to contribute no significant amount of activation, so mismatch similarity plays no role). We found these six cases of parameter values varying with the condition to be a minimal set needed to produce an acceptable model, though of course only minimal with respect to the very small set of model variations we were able to pursue and explore over the several years trying to come up with an acceptable model. Scientists might well disagree about the 'reasonableness' of such similarity differences but they have proved critical in other domains, notably the exemplar modeling of Nosofsky (and others) for categorization.

Conclusion

This research presents a comprehensive model of the processes used in short-term recognition memory. It contains two key elements governing recognition decisions: the joint use of familiarity and learning of responses. These are based on storage in and retrieval from short-term memory, long-term event memory (i.e., memory traces from prior trials), and long-term knowledge (encoding response learning).

The present data and modeling have shown that there exists a model comprised of what we believe to be the important core processes of storage in and retrieval from short-term memory, long-term event memory, and long-term knowledge, a model able to capture all the patterns of data in several studies of short-term recognition. The processes in question have been supported in numerous research studies, some going back to the first days of experimental psychology and others relatively recent, but have not previously been shown to operate together to predict such a wide array of experimental conditions with measures of both accuracy and response time. The three studies and nine groups measured both accuracy and response time in conditions that varied (a) target versus foil; (b) list lengths from 2 to 9; (c) target position in the list; (d) trials using varied mapping (VM), in which targets and foils changed roles from trial to trial (preventing learning); versus trials using consistent mapping (CM), in which in the standard paradigm targets and foils retained those roles on every trial (allowing learning of responses); versus trials using a new kind of consistent mapping in a new paradigm in which each picture served as both target and foil and was equally familiar but allowed learning of the response for each picture; (e) pure conditions in which all trials were VM, AN, or both kinds of CM versus Mixed conditions in which each trial contained two or three of these trial types; (f) centered presentations (in Nosofsky et al. 2021), versus left or right side presentations (in the present two experiments). Such a wide array of conditions provides a strong test of the proposed memory processes, and the successful application of the model gives support to their importance and the way they are posited to work together.

Data Availability Data can be downloaded from the Open Science Framework (OSF) at <https://osf.io/jp7b3/>.

Code Availability The Julia program scripts used for modeling and the Matlab program scripts for creating experiments are available for download on OSF at <https://osf.io/jp7b3/>.

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